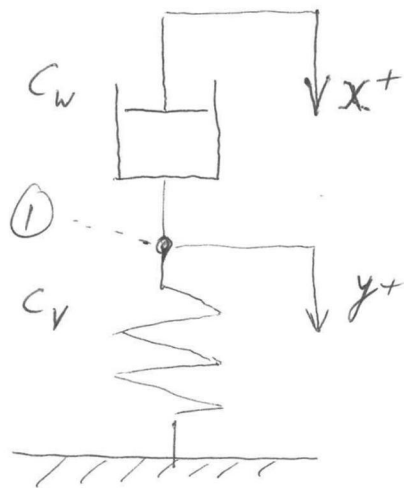


§ 2.9 opgaven

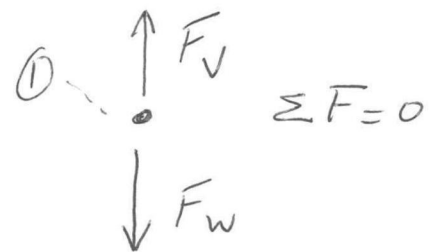
6c)



Bereken de krachtenbalans in punt "1"

a.) $F_{veer} = F_V = c_V \cdot y(t)$

$$F_{dempert} = F_W = c_W \cdot v(t) \\ = c_W \cdot \frac{d(y(t) - x(t))}{dt}$$



$$\Sigma F = 0 \rightarrow F_V - F_W = 0$$

invullen: $c_V \cdot y(t) - c_W \frac{d(y(t) - x(t))}{dt} + c_W \cdot \frac{dy(t)}{dt} = 0$

b.) naar s-domein: $c_V \cdot y(s) - c_W \cdot s \cdot X(s) + c_W \cdot s \cdot y(s) = 0$

$$c_V \cdot y(s) + c_W \cdot s \cdot y(s) = c_W \cdot s \cdot X(s)$$

$$y(s) \cdot (c_V + c_W \cdot s) = c_W \cdot s \cdot X(s)$$

$$H(s) = \frac{y(s)}{X(s)} = \frac{c_W \cdot s}{(c_V + c_W \cdot s)} = \frac{\frac{c_W}{c_V} \cdot s}{\frac{c_V}{c_V} + \frac{c_W \cdot s}{c_V}} = \frac{\frac{c_W}{c_V} \cdot s}{\frac{c_W}{c_V} \cdot s + 1}$$